FACULTY OF COMPUTING
UTM Johor Bahru

SCST1223: PROBABILITY & STATISTICAL DATA ANALYSIS
2025/2026 – SEMESTER 2

PROJECT PHASE 1 : Data Collection and Descriptive Analysis

Objective: The primary aim of Project Phase 1 is to provide hands-on experience in **primary data collection** and **descriptive statistical analysis**.

Analysis Requirements: The analysis for this phase is restricted to **descriptive statistics**. Your report should include:

- **Frequency Distributions** : Summarizing the raw data.
- **Categorical Data Visualization** : Utilizing **bar charts** or pie charts.
- **Numerical Data Visualization** : Utilizing **stem-and-leaf plots** or histograms.
- **Measures of Central Tendency** : Calculating the **mean, median, and mode**.
- **Measures of Dispersion** : (Optional but recommended) Calculating the range and standard deviation.

Group Formation and Scope: This project must be conducted in groups of **no more than five (5) students**. Each group should collaborate to identify a specific research topic within the assigned industry scope:

1. **Section 01:** Healthcare

Students are required to conduct original data collection for this project, **ensuring a minimum sample size of 60 respondents**. Data must be gathered using a structured questionnaire, which must be included as an appendix in the final report. All data analysis is to be performed using R programming. The results and interpretation must be compiled into a formal report, organized according to the following structure:

1) **Introduction or background**
   Explain the purpose of the study in your own words, why you were interested in the question and what you expected to see from the data.
2) **Data Collection**
   Data description including parameters or variables to be measured.
3) **Data Analysis**
   a. Categorical data:
      - **Graphs**: Pie chart or bar chart (with percentages)
      - **Numerical**: Proportion and sample size in each category
   b. Quantitative data:
      - **Graphs**: E.g., histograms or dot plots.
      - **Measures of Central Tendency**: E.g., mean, media, boxplots

   Interpret and discuss what you can learn from the pictures (e.g., compare shape…) and make sure you report the sample sizes for the survey as well.
4) **Conclusion**
   Write a one-paragraph summary on what you learned about the population you surveyed. Explain what you learn mathematically and/or technologically from conducting this study? Any final thoughts.

Report must be submitted through https://elearning.utm.my/25262 on/before **Week 7 (26 April 2026)**.